Impossibility Result of Local Linear Key Distillation for Algebraic Authentication Codes with Arbitration

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Abstract: An authentication code with arbitration (A^2-code) is an information-theoretically secure cryptographic technique for message authentication when the sender and the receiver do not fully trust each other. A trusted arbiter helps resolve disputes and detect cheating by either party. Such techniques are usually described using uniformly random keys. In practice, however, randomness may be biased or partially leaked, and may only be known to have a certain amount of min-entropy. Such randomness is commonly modeled as a weak source. In this paper, we study how to transform the correlated keys of a finite-field algebraic A^2-code generated from weak sources into keys close to the ideal distribution obtained from uniform randomness. A simple approach is to use a public seed and have the sender and receiver apply local linear transformations to their respective key shares. We show that this approach does not work if the algebraic relation among the keys must be preserved exactly. In particular, public-seeded local linear transformations cannot both preserve the required relation and produce keys close to the ideal distribution. This shows that key distillation for this A^2-code requires more than straightforward local linear extraction. We also revisit the key distillation protocol presented at CSS 2025. The local transformations used in the earlier presentation do not preserve the required key relation. We therefore give a corrected protocol in which the necessary correction information is sent over a private channel.

Keywords: Authentication code with arbitration, Weak source, Key distillation, Universal hashing, Information-theoretic security.

1. Introduction

1.1 Background and Motivation

An authentication code (A-code) is a cryptographic technique that allows a receiver to detect if a message has been tampered with by an external adversary. The pioneering research on this technique was conducted by Gilbert et al. [4] and formalized within game-theoretic and information-theoretic frameworks by Simmons [10]. Under the A-code setting, a sender and a receiver maintain mutual trust and collaborate to protect the message against adversarial attacks.

Authentication codes with arbitration (A^2-codes) were introduced by Simmons to address a setting in which the sender and the receiver do not necessarily trust each other [11]. An A^2-code introduces a trusted arbiter who holds information related to both parties' secret keys and resolves disputes between them. It therefore provides protection not only against an external opponent,but also against dishonest behavior by the sender or the receiver, such as repudiating an honestly generated message or claiming receipt of a message that was never sent.

Standard information-theoretic cryptography assumes uniformly random secret keys. However, real-world secrets may be biased or partially leaked, and may retain only conditional min-entropy. Dodis and Wichs showed that two mutually trusting parties sharing a common weak secret can perform message authentication and authenticated key agreement over an unauthenticated public channel [3]. Their setting, however, does not address authentication with arbitration, in which the sender and receiver may be dishonest and hold distinct, correlated keys distributed by an arbiter.

Moreover, the obstacle is not merely that the two parties hold different weak sources. The sender and receiver keys of an A^2-code must satisfy algebraic relations required for receiver verification and dispute resolution. Independently extracting the two keys may destroy these relations. On the other hand, publicly sharing the seed and applying seeded functions to the key remains a desirable solution, as it avoids sharing key-dependent values. This leads to the following question: *Can the correlated weak keys of an algebraic A^2-code be locally distilled toward the ideal distribution while exactly preserving the algebraic relations required for verification and arbitration?*

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1.2 Our Contribution

In this work, we focus on the class of local linear transformations. We then investigate whether the correlated weak keys can be distilled by applying seeded linear transformations locally and independently at the sender and the receiver. We prove that such a transformation cannot, in general, simultaneously preserve the algebraic relations required by the underlying A^2-code and produce a key pair close to the corresponding ideal uniformly distributed keys. This establishes a limitation of the most direct extraction-based approach.

Finally, we give a corrected construction that circumvents the above limitation by combining component-wise linear universal hashing with authenticated transmission of correction terms. Message authentication from weak sources is used to authenticate a common hash function, while authenticated key agreement from weak sources and information-theoretic one-time authenticated encryption are used to protect the correction terms. We also refine the security model used at CSS 2025 by evaluating each deception probability conditioned on the attacker's view, including the public transcript, rather than only on the attacker's key. The distilled keys preserve the algebraic relations of the underlying A^2-code and can therefore be used with its standard message authentication and verification algorithms denoted by MAC and Vf, respectively.

2. Preliminaries

2.1 Notations

In this paper, unless otherwise specified, random variables are denoted by uppercase letters (e.g., X), and their realizations by lowercase letters (e.g., x). In particular, a random variable uniformly distributed on the set $\mathcal{X}$ is denoted by $U_{\mathcal{X}}$. Furthermore, the operation of uniformly randomly selecting an element x from the set $\mathcal{X}$ is denoted by $x \leftarrow\!\$\, \mathcal{X}$. These notations are used appropriately as needed. Regarding the notation for probability, we may abbreviate $\Pr[X = x] = P_X(x)$ as needed; in particular, for a uniform distribution, we may write $\Pr[U_{\mathcal{X}} = x] = \Pr_{x \leftarrow\!\$ \mathcal{X}}[x]$. The expected value of a random variable X is denoted by $\mathbb{E}_X[\cdot]$. Furthermore, the support $\{x \in \mathcal{X} : \Pr[X = x] > 0\}$ of a random variable X defined on the set $\mathcal{X}$ is denoted by $\mathrm{Supp}(X)$. The output y of a probabilistic algorithm A that takes x as input is denoted by $y \leftarrow A(x)$. All logarithms are taken to base 2.

2.2 Weak Sources and Statistical Distance

Standard A-code constructions typically assume access to uniformly distributed secret keys. Instead, we consider keys generated from non-uniform sources whose exact distributions may be unknown, but whose min-entropy is guaranteed to be sufficiently large. We use min-entropy and average conditional min-entropy to quantify their unpredictability, and statistical distance to measure closeness to the uniform distribution.

The min-entropy and average-case conditional min-entropy of random variables are defined as follows.

Definition 1 (Min-entropy)**.** *The min-entropy of a random variable X over $\mathcal{X}$ is*

$$\mathrm{H}_\infty(X) \coloneqq \min_{x \in \mathcal{X}} \{-\log \Pr[X = x]\}.$$

Using min-entropy, a k-source is defined as follows.

Definition 2 (k-source)**.** *A random variable X is a k-source, if $\mathrm{H}_\infty(X) \geq k$.*

Definition 3 (Average-case conditional min-entropy [1])**.** *The average-case conditional min-entropy of X given Z is defined as*

$$\widetilde{\mathrm{H}}_\infty(X \mid Z) \coloneqq -\log\left(\mathbb{E}_Z\left[\max_x \Pr[X = x \mid Z = z]\right]\right).$$

Equivalently,

$$\widetilde{\mathrm{H}}_\infty(X \mid Z) \coloneqq -\log P_{\mathrm{guess}}(X \mid Z),$$

where

$$P_{\mathrm{guess}}(X \mid Z) \coloneqq \mathbb{E}_Z[\max_x P_{X|Z}(x \mid z)].$$

Min-entropy measures the unpredictability of a random variable, but it does not directly quantify how close its distribution is to the uniform distribution. To measure the difference between probability distributions, we use statistical distance. In particular, statistical distance allows us to express that a random variable obtained from a weak source is almost uniformly distributed, possibly even in the presence of side information.

A statistical distance between random variables is defined as follows.

Definition 4 (Statistical distance)**.** *Let X_1, X_2 be random variables over $\mathcal{X}$. Then a statistical distance $\Delta(X_1; X_2)$ between X_1 and X_2 is*

$$\Delta(X_1; X_2) \coloneqq \frac{1}{2} \sum_{x \in \mathcal{X}} |\Pr[X_1 = x] - \Pr[X_2 = x]|.$$

For a random variable Y over $\mathcal{Y}$, the conditional statistical distance between X_1 and X_2 conditioned on Y is

$$\Delta(X_1; X_2 \mid Y) \coloneqq \frac{1}{2}\,\mathbb{E}_Y\left[\sum_{x \in \mathcal{X}} \left|\Pr[X_1 = x \mid Y] - \Pr[X_2 = x \mid Y]\right|\right].$$

Here, it can be shown that $\Delta(X_1; X_2 \mid Y) = \Delta((X_1, Y); (X_2, Y))$.

Furthermore, a statistical distance has the following property [12]:

Proposition 1 (A monotonicity of a statistical distance)**.** *Let X_1, X_2 be random variables over $\mathcal{X}$. For any finite sets $\mathcal{Y}$ and any functions $f : \mathcal{X} \to \mathcal{Y}$,*

$$\Delta(X_1; X_2) \geq \Delta(f(X_1); f(X_2)).$$

Universal hashing defined by Definition 5 provides a standard method for extracting almost-uniform randomness from a source with sufficiently large min-entropy. This fact

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is known as the leftover hash lemma [5]. In this work, we require its generalized form, as defined in Proposition 2, in the presence of correlated side information. We therefore state only the following version, formulated in terms of average conditional min-entropy.

Definition 5. *A function family $\mathcal{H} \coloneqq \{h : \mathcal{X} \to \mathcal{Y}\}$ is called universal if*

$$\Pr_{h \leftarrow\$ \mathcal{H}}[h(x_1) = h(x_2)] \leq \frac{1}{|\mathcal{Y}|},$$

for any distinct $x_1, x_2 \in \mathcal{X}$.

Definition 6. *A function family $\mathcal{H} \coloneqq \{h : \mathbb{F}_q \to \mathbb{F}_r\}$ is called $\mathbb{F}_2$-linear universal if every $h \in \mathcal{H}$ is $\mathbb{F}_2$-linear and $\mathcal{H}$ is a universal family.*

Proposition 2 (Generalized leftover hash lemma [2])**.** *Let (X, Z) be jointly distributed random variables such that*

$$\widetilde{\mathrm{H}}_\infty(X \mid Z) \geq k.$$

Let $\mathcal{H}$ be a universal family of hash functions from $\mathcal{X}$ to $\mathcal{Y}$, and let H be sampled uniformly from $\mathcal{H}$ independently of (X, Z). Then

$$\Delta\left(H(X); U_{\mathcal{Y}} \mid H, Z\right) \leq \frac{1}{2}\sqrt{|\mathcal{Y}|\, 2^{-k}}.$$

2.3 Algebraic Authentication Code with Arbitration

An authentication code with an arbiter, or an A^2-code, is a message authentication scheme designed for a setting in which not only an external opponent but also the sender or the receiver may be dishonest. The model consists of four parties: a sender, a receiver, an opponent, and an arbiter. The arbiter is assumed to be honest and resolves disputes between the sender and the receiver.

We formulate the model following the standard model of A^2-codes introduced by Johansson [6], while reorganizing its presentation to facilitate the extension to weak-source keys introduced later. In particular, we describe the protocol in terms of the algorithms MAC and Vf and explicitly separate the key-generation, key-distribution, message-generation, receiver-verification, and arbitration phases.

We focus on A^2-codes without message secrecy. (Such an A^2-code is called a Cartesian A^2-code.) Thus, the purpose of the code is to provide message authentication and arbitration, but not secrecy. An authenticated message is represented as (m, t), where $m \in \mathcal{M}$ is transmitted in plaintext and $t \in \mathcal{T}$ is its authentication tag. The sender and the receiver communicate over a public unauthenticated channel; hence, the opponent may observe, modify, delete, and inject authenticated messages.

An A^2-code consists of the following algorithms:

$$\mathsf{MAC}\colon \mathcal{M} \times \mathcal{K}_{\mathrm{S}} \longrightarrow \mathcal{T},$$
$$\mathsf{Vf}\colon \mathcal{M} \times \mathcal{T} \times \mathcal{K}_{\mathrm{R}} \longrightarrow \{0, 1\}.$$

The arbiter resolves a dispute using $\mathsf{MAC}, \mathsf{Vf}$ and both parties' keys.

The protocol consists of the following phases.

Key-generation phase. The arbiter samples a pair of keys $(K_{\mathrm{S}}, K_{\mathrm{R}})$ from $\mathcal{K}_{\mathrm{S}} \times \mathcal{K}_{\mathrm{R}}$ according to the key distribution specified by the A^2-code.

Key-distribution phase. The arbiter sends K_{S} to the sender and K_{R} to the receiver over private authenticated channels. The arbiter retains both keys for possible dispute resolution.

Message-generation phase. To authenticate a message $m \in \mathcal{M}$, the sender computes

$$t \leftarrow \mathsf{MAC}(m, K_{\mathrm{S}})$$

and sends the authenticated message (m, t) to the receiver over a public unauthenticated channel.

Receiver-verification phase. Upon receiving an authenticated message $(\tilde{m}, \tilde{t})$, the receiver computes

$$d_{\mathrm{R}} \leftarrow \mathsf{Vf}(\tilde{m}, \tilde{t}, K_{\mathrm{R}}).$$

The receiver accepts $\tilde{m}$ if $d_{\mathrm{R}} = 1$ and rejects $(\tilde{m}, \tilde{t})$ otherwise.

Arbitration phase. If a dispute arises regarding a pair of message and tag (m, t), the arbiter confirms whether (m, t) can be generated from MAC by using $\mathsf{MAC}, \mathsf{Vf}, k_{\mathrm{S}}, k_{\mathrm{R}}$.

Definition 7 (Correctness of an A^2-code)**.** *An A^2-code is correct if, for every key pair $(k_{\mathrm{S}}, k_{\mathrm{R}}) \in \mathrm{Supp}(K_{\mathrm{S}}, K_{\mathrm{R}})$ and every message $m \in \mathcal{M}$, an honestly generated authenticated message is accepted by the receiver and attributed to the sender by the arbiter. That is,*

$$\mathsf{Vf}(m, \mathsf{MAC}(m, k_{\mathrm{S}}), k_{\mathrm{R}}) = 1.$$

The standard security model considers five deception attacks: impersonation (I) and substitution (S) by an external opponent, impersonation by the sender (T), and impersonation (R_0) and substitution (R_1) by the receiver. We refer to [6], [7] for their formal definitions. For each attack $\mathrm{A} \in \{I, S, T, R_0, R_1\}$, let $\mathsf{View}_{\mathrm{A}}$ denote the attacker's view and $\mathsf{Succ}_{\mathrm{A}}$ the event that the attack succeeds, and define

$$P_{\mathrm{A}} \coloneqq \mathbb{E}_{\mathsf{View}_{\mathrm{A}}} \Pr[\mathsf{Succ}_{\mathrm{A}} \mid \mathsf{View}_{\mathrm{A}}].$$

In this paper, we evaluate the attack probability of each attack conditioned on the attacker's view. The view includes not only the attacker's own key but also local states, local outputs, and public information obtained during protocol execution, such as the transcript of the key-distillation procedure. This differs from our formulation at CSS 2025, in which the attack probabilities were conditioned only on the attacker's secret key.

Definition 8 (Security of an A^2-code)**.** *The deception probability of an A^2-code is defined as*

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Algorithm 1 A^2-code protocol [7]

Require: Message space $\mathcal{M}$, tag space $\mathcal{T}$

Phase 1: Key generation and distribution

Arbiter samples $e_1, e_2, f_1, f_2, f_3 \leftarrow_{\$} \mathbb{F}_p$.

Arbiter computes

$$e_3 \coloneqq f_1 + e_1 f_3,$$
$$e_4 \coloneqq f_2 + e_2 f_3.$$

Arbiter sets $k_S \leftarrow (e_1, e_2, e_3, e_4)$.

Arbiter sets $k_R \leftarrow (f_1, f_2, f_3)$.

Arbiter sends k_S to Sender over an authenticated private channel.

Arbiter sends k_R to Receiver over an authenticated private channel.

Phase 2: Sending message and authentication

Sender chooses $m \in \mathcal{M}$.

Sender computes $(t_1, t_2) \leftarrow \mathsf{MAC}(m, k_S)$ where

$$t_1 \coloneqq e_1 + m e_2,$$
$$t_2 \coloneqq e_3 + m e_4.$$

Sender sends (m, t_1, t_2) to Receiver over an unauthenticated public channel.

Receiver receives $(\tilde{m}, \tilde{t_1}, \tilde{t_2})$.

if $\tilde{t_2} = f_1 + \tilde{m} f_2 + \tilde{t_1} f_3$ **then**

    Receiver accepts $\tilde{m}$.

else

    Receiver rejects $\tilde{m}$.

end if

Phase 3: Dispute resolution

Suppose a dispute occurs for a claimed tuple $(\tilde{m}, \tilde{t_1}, \tilde{t_2})$.

Arbiter recomputes $k_S = (e_1, e_2, e_3, e_4)$ and $k_R = (f_1, f_2, f_3)$.

if $\mathsf{MAC}(\tilde{m}, k_S) = (t_1, t_2)$ **then**

    The arbiter accepts $\tilde{m}$.

else

    The arbiter rejects $\tilde{m}$.

end if

function $\mathsf{MAC}(m, (e_1, e_2, e_3, e_4))$

    $t_1 \coloneqq e_1 + m e_2$

    $t_2 \coloneqq e_3 + m e_4$

    return (t_1, t_2)

end function

function $\mathsf{Vf}(m, (t_1, t_2), (f_1, f_2, f_3))$

    if $t_2 = f_1 + m f_2 + t_1 f_3$ **then**

        return 1

    else

        return 0

    end if

end function

$$P_D = \max\{P_I, P_S, P_T, P_{R_0}, P_{R_1}\}.$$

An A^2-code is ε-secure if $P_D \le \varepsilon$.

In this paper, we use a standard algebraic construction of Cartesian A^2-codes due to Johansson. [7], which is described as Algorithm 1.

3. Revisiting the Protocol of CSS 2025

3.1 Local Key Transformation Proposed at CSS 2025

In CSS 2025 [8], we studied the same problem of instantiating the algebraic A^2-code of Algorithm 1 from correlated weak-source keys, and proposed a two-stage local hashing procedure. There, the key pair (k_S, k_R) is sampled from a k-source over $\mathbb{F}_p^4 \times \mathbb{F}_p^3$ for a prime p, so that the raw key components $e_1, e_2, e_3, e_4, f_1, f_2, f_3 \in \mathbb{F}_p$ satisfy

$$e_3 = f_1 + e_1 f_3, \tag{1}$$
$$e_4 = f_2 + e_2 f_3, \tag{2}$$

computed in $\mathbb{F}_p$, where e_1, e_2, f_1, f_2, f_3 are mutually independent k-sources over $\mathbb{F}_p$. To obtain keys usable with an A^2-code of the form of Algorithm 1 over a much smaller field, the construction of [8] compresses the raw keys in two stages, through an intermediate field $\mathbb{F}_{q_1}$ and a final field $\mathbb{F}_{q_2}$, for primes $p > q_1 > q_2$. [*1]

The sender and the receiver first obtain four common seed values $a, b \in \mathbb{F}_{q_1}$ and $a', b' \in \mathbb{F}_{q_2}$: the sender shares a, b, a', b' with the arbiter using the A-code from weak sources of Proposition 3, and the arbiter then relays them to the receiver using the same primitive. Both parties then apply the following seeded transformation locally.

Sender. The sender computes

$$e_i' \coloneqq a e_i + b \bmod q_1, \qquad i = 1, 2, 3, 4,$$
$$e_i'' \coloneqq e_i' \bmod q_2, \qquad i = 1, 2,$$
$$e_j'' \coloneqq a' e_j' + b'(e_{j-2}' + 1) \bmod q_2, \qquad j = 3, 4,$$

and sets $k_S' \coloneqq (e_1'', e_2'', e_3'', e_4'')$.

Receiver. The receiver computes

$$f_i' \coloneqq a f_i - b(f_3 - 1) \bmod q_1, \qquad i = 1, 2,$$
$$f_3' \coloneqq f_3 \bmod q_1,$$
$$f_i'' \coloneqq a' f_i' + b' \bmod q_2, \qquad i = 1, 2, 3,$$

and sets $k_R' \coloneqq (f_1'', f_2'', f_3'')$.

The pair (k_S', k_R') is then used with $\mathsf{MAC}, \mathsf{Vf}$ of Algorithm 1 over $\mathbb{F}_{q_2}$. [*2]

3.2 Failure to Preserve the Algebraic Relation

We now show that the transformation of Section 3.1 does not preserve Eqs. (3) and (4), the algebraic relation required by Algorithm 1. We first isolate the reason the construction

[*1] We wrote q_1, q_2 for the two field sizes used in [8], where we denoted them by q, r, reserving q for the base-field size of $\mathbb{F}_{q^n}$ used from Section 4 onward.

[*2] We state the receiver-side transformation with the sign $-b(f_3 - 1)$, rather than $+b(f_3 - 1)$ as it appears in [8]; Lemma 1 shows that this is the sign for which the sender- and receiver-side transformations are mutually consistent in the absence of modular reduction, so we analyze the construction in this, its intended, form.

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was nonetheless plausible: if no modular reduction were applied, i.e., if e_i', e_i'', f_i', f_i'' were defined by the same expressions as in Section 3.1 but evaluated as exact integer (or rational) expressions rather than reduced modulo q_1 or q_2, the relation would hold exactly.

Lemma 1. *Suppose $e_3 = f_1 + e_1 f_3$ and $e_4 = f_2 + e_2 f_3$ hold as exact identities, and define e_i', e_i'', f_i', f_i'' by the formulas of Section 3.1 without reducing modulo q_1 or q_2. Then,*

$$e_3'' = f_1'' + e_1'' f_3'',$$
$$e_4'' = f_2'' + e_2'' f_3''.$$

Proof. We verify the first relation; the second follows identically. By construction, $e_1' = ae_1 + b$, $e_3' = ae_3 + b$, $f_1' = af_1 - b(f_3 - 1)$, and $f_3' = f_3$. Using $e_3 = f_1 + e_1 f_3$, it holds that

$$\begin{aligned} f_1' + e_1' f_3' &= af_1 - b(f_3 - 1) + (ae_1 + b)f_3 \\ &= a(f_1 + e_1 f_3) - bf_3 + b + bf_3 \\ &= ae_3 + b = e_3'. \end{aligned}$$

Since $e_1'' = e_1'$ and $f_3'' = a' f_3' + b'$, $f_1'' = a' f_1' + b'$, and $e_3'' = a' e_3' + b'(e_1' + 1)$, we have

$$\begin{aligned} f_1'' + e_1'' f_3'' &= a' f_1' + b' + e_1'(a' f_3' + b') \\ &= a'(f_1' + e_1' f_3') + b'(1 + e_1') \\ &= a' e_3' + b'(e_1' + 1) = e_3''. \end{aligned}$$ □

Lemma 1 shows that the receiver's side correction term $-b(f_3 - 1)$ and the sender's side correction term $b'(e_1' + 1)$ were designed precisely to cancel the discrepancy that the multiplicative term $e_1 f_3$ would otherwise introduce, and that they do so exactly, provided that no modular reduction is applied. In the actual protocol, however, the key components are compressed across finite fields of different characteristics. In particular, $e_i', f_i' \in \mathbb{F}_{q_1}$ must be further compressed into $\mathbb{F}_{q_2}$ so that the resulting keys are close to uniform over the working field of the underlying A^2-code. This compression is necessary for randomness extraction. Since q_1 and q_2 are distinct primes, $\mathbb{F}_{q_1}$ and $\mathbb{F}_{q_2}$ have different characteristic, and there is no ring homomorphism $\mathbb{F}_{q_1} \to \mathbb{F}_{q_2}$. Consequently, the map $x \mapsto x \bmod q_2$ applied to an integer representative $x \in \{0, \ldots, q_1 - 1\}$ of an element of $\mathbb{F}_{q_1}$ is not compatible with the field operations of $\mathbb{F}_{q_1}$: for $e_1' \in \mathbb{F}_{q_1}$, the value $e_1'' = e_1' \bmod q_2$ need not agree with $(ae_1 + b) \bmod q_2$ computed directly from the unreduced expression, so the cancellation established in Lemma 1 over $\mathbb{F}_{q_1}$ is not preserved once the keys are reduced into $\mathbb{F}_{q_2}$.

The following numerical example exhibits this failure concretely. Let $p = 101$, $q_1 = 17$, $q_2 = 5$, and let $a = 1, b = 15 \in \mathbb{F}_{q_1}$ and $a' = 3, b' = 1 \in \mathbb{F}_{q_2}$ be the shared seeds. Take raw keys $e_1 = 21, f_1 = 87, f_3 = 94 \in \mathbb{F}_p$ and set $e_3 \coloneqq (f_1 + e_1 f_3) \bmod p = 41$, so that Eq. (1) holds. Applying the transformation of Section 3.1 gives $e_1' = 2$, $f_1' = 1$, $f_3' = 9$, $e_3' = 5$ in $\mathbb{F}_{q_1}$, and hence $e_1'' = 2$, $f_1'' = 4$, $f_3'' = 3$, $e_3'' = 3$ in $\mathbb{F}_{q_2}$. However,

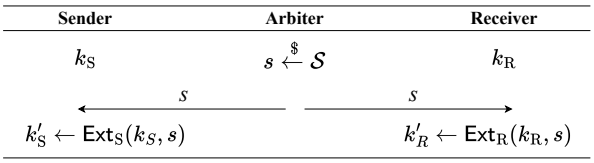


Fig. 1 Seeded local key transformation followed by an A^2-code

$$f_1'' + e_1'' f_3'' = 4 + 2 \cdot 3 = 10 \equiv 0 \not\equiv 3 = e_3'' \pmod{q_2}.$$

Even when the transformation of Section 3.1 is applied to keys in Eq. (2), the algebraic relation can similarly break down. Thus, the transformation of [8] does not preserve the algebraic relation required by Algorithm 1, even though the correction terms are designed so that the relation holds exactly whenever no modular reduction is applied. This failure raises a more general question: whether a different choice of public-seeded local linear transformations could preserve the algebraic relation while distilling the keys toward the ideal distribution. Section 4 shows that, even within this natural class, the two requirements cannot generally be satisfied simultaneously.

4. Impossibility of Local Linear Key Distillation

4.1 Local Linear Key Distillation

In this paper, we study A^2-codes in which the sender and receiver keys are generated from k-sources. A natural approach is to first share a public seed over an unauthenticated channel using the A-code from weak sources (shown in Fig. 1), and then apply seeded extractors locally to the sender and receiver keys of Algorithm 1. However, applying extractors independently to the two keys generally destroys the algebraic relation (shown in Algorithm 1 line 2) required by the A^2-code. It is therefore natural to ask whether seed-dependent linear transformations can simultaneously preserve this relation and convert the original weak keys into nearly uniform keys.

Theorem 1 in this section shows that, in general, no such linear transformations exist. This means that in the key-distillation phase, a simple key distillation strategy, in which the sender ' s key and the receiver ' s key are each independently input into a linear extractor using a public seed, while still strictly preserving the key correlation of the A^2-code, is not possible.

4.2 Impossibility Theorem

The image-size bound derived below implies a lower bound on the statistical distance between the transformed receiver key and the ideal receiver key for every fixed seed. Finally, the statistical distance conditioned on the public seed can be written as

$$\Delta\big((S, K_R'), (S, \widehat{K}_R)\big) = \mathbb{E}_{s \sim S}\left[\Delta\left(K_R'; \widehat{K}_R \mid S = s\right)\right].$$

Therefore, a lower bound that holds for every fixed seed immediately gives the same lower bound on the seed-conditional statistical distance.

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Theorem 1. *Let $(K_{\mathrm{S}}, K_{\mathrm{R}})$ be an arbitrarily distributed random key pair supported on key pairs satisfying*

$$e_3 = f_1 + e_1 f_3,$$
$$e_4 = f_2 + e_2 f_3.$$

Let S be a random variable over the seed space $\mathcal{S}$. For any $n > \ell$, there exist no $\mathbb{F}_q$-linear seeded maps

$$\mathsf{Ext}_{\mathrm{S}} : \mathbb{F}_{q^n}^4 \times \mathcal{S} \to \mathbb{F}_{q^\ell}^4,$$
$$\mathsf{Ext}_{\mathrm{R}} : \mathbb{F}_{q^n}^3 \times \mathcal{S} \to \mathbb{F}_{q^\ell}^3$$

such that, defining

$$K'_{\mathrm{S}} \coloneqq \mathsf{Ext}_{\mathrm{S}}(K_{\mathrm{S}}, S),$$
$$K'_{\mathrm{R}} \coloneqq \mathsf{Ext}_{\mathrm{R}}(K_{\mathrm{R}}, S),$$

both

$$\Delta\left((S, K'_{\mathrm{S}}, K'_{\mathrm{R}}); (S, \widehat{K_{\mathrm{S}}}, \widehat{K_{\mathrm{R}}})\right) < 1 - q^{-2\ell}$$

and

$$e'_3 = f'_1 + e'_1 f'_3, \tag{3}$$
$$e'_4 = f'_2 + e'_2 f'_3, \tag{4}$$

hold for every $s \in \mathcal{S}$ and every key pair satisfying the above relations, where

$$k'_{\mathrm{S}} \coloneqq (e'_1, e'_2, e'_3, e'_4) \coloneqq \mathsf{Ext}_{\mathrm{S}}((e_1, e_2, e_3, e_4), s),$$
$$k'_{\mathrm{R}} \coloneqq (f'_1, f'_2, f'_3) \coloneqq \mathsf{Ext}_{\mathrm{R}}((f_1, f_2, f_3), s). \tag{5}$$

Here, $(\widehat{K_{\mathrm{S}}}, \widehat{K_{\mathrm{R}}})$ denotes a uniformly distributed pair of keys satisfying Eqs. (3) and (4), independently of S.

4.3 Proof of the Impossibility Theorem

Theorem 1. We will prove that

$$\Delta\left((S, K'_{\mathrm{R}}); S, \widehat{K_{\mathrm{R}}}\right) \geq 1 - q^{-2\ell},$$

because Proposition 1 yields

$$\Delta\left((S, K'_{\mathrm{S}}, K'_{\mathrm{R}}); S, \widehat{K_{\mathrm{S}}}, \widehat{K_{\mathrm{R}}}\right) \geq \Delta\left((S, K'_{\mathrm{R}}); (S, \widehat{K_{\mathrm{R}}})\right).$$

At first, fix a seed of $\mathsf{Ext}_{\mathrm{S}}$ and $\mathsf{Ext}_{\mathrm{R}}$ by s, and define $\phi_{\mathrm{S}}(\cdot) = \mathsf{Ext}_{\mathrm{S}}(\cdot, s), \phi_{\mathrm{R}}(\cdot) = \mathsf{Ext}_{\mathrm{R}}(\cdot, s)$, we fix a sender key $K_{\mathrm{S}} = k_{\mathrm{S}} = (e_1, e_2, 0, 0)$. Then, the set of receiver keys consistent with this k_{S} is

$$\mathcal{L}_{\mathrm{R}}(e_1, e_2) \coloneqq \{(-e_1 t, -e_2 t, t) \colon t \in \mathbb{F}_{q^n}\}.$$

Now, a receiver key $k_{\mathrm{R}} = (0, 0, 0)$ is an element of $\mathcal{L}_{\mathrm{R}}(e_1, e_2)$. By linearity of ϕ_{R} and Eq. (5), we obtain

$$(f'_1, f'_2, f'_3) = \mathsf{Ext}_{\mathrm{R}}((0, 0, 0), s) = (0, 0, 0),$$

which yields from Eqs. (3) and (4) that

$$e'_3 = 0 + e'_1 \cdot 0 = 0,$$
$$e'_4 = 0 + e'_2 \cdot 0 = 0.$$

Here, the output (e'_1, e'_2, e'_3, e'_4) is independent of (f_1, f_2, f_3); rather, it depends only on (e_1, e_2, e_3, e_4) and s. So for any (f_1, f_2, f_3),

$$\phi_{\mathrm{S}}(e_1, e_2, 0, 0) = (e'_1, e'_2, 0, 0). \tag{6}$$

From Eqs. (3), (4) and (6), for arbitrary receiver key $(-e_1 t, -e_2 t, t) \in \mathcal{L}_{\mathrm{R}}(e_1, e_2)$, there exists $u \in \mathbb{F}_{q^l}$,

$$\phi_{\mathrm{R}}(-e_1 t, -e_2 t, t) = (-e'_1 u, -e'_2 u, u).$$

Here, define the set

$$\mathcal{L}'_{\mathrm{R}}(e'_1, e'_2) \coloneqq \{(-e'_1 u, -e'_2 u, u) \colon u \in \mathbb{F}_{q^\ell}\}.$$

Then,

$$\phi_{\mathrm{R}}(\mathcal{L}_{\mathrm{R}}(e_1, e_2)) \subseteq \mathcal{L}'_{\mathrm{R}}(e'_1, e'_2). \tag{7}$$

We note that $\mathcal{L}_{\mathrm{R}}(e_1, e_2)$ is a linear subspace of $\mathbb{F}_{q^n}$ and $\mathcal{L}'_{\mathrm{R}}(e'_1, e'_2)$ is a linear subspace of $\mathbb{F}_{q^\ell}$.

Next, we consider the kernel of $\phi_{\mathrm{R}} : \mathcal{L}_{\mathrm{R}}(e_1, e_2) \to \mathcal{L}'_{\mathrm{R}}(e'_1, e'_2)$ to evaluate the kernel of $\phi_{\mathrm{R}} : \mathbb{F}_{q^n}^3 \to \mathbb{F}_{q^\ell}^3$. We denote the restriction $\phi_{\mathrm{R}} : \mathcal{L}_{\mathrm{R}}(e_1, e_2) \to \mathcal{L}'_{\mathrm{R}}(e'_1, e'_2)$ by $\phi_{\mathrm{R}}|_{\mathcal{L}_{\mathrm{R}}(e_1, e_2)}$. Now, we obtain

$$\dim_{\mathbb{F}_q} \mathcal{L}_{\mathrm{R}}(e_1, e_2) = n,$$
$$\dim_{\mathbb{F}_q} \mathcal{L}'_{\mathrm{R}}(e'_1, e'_2) = \ell. \tag{8}$$

Here, we identify $\mathbb{F}_{q^n}$ and $\mathbb{F}_{q^\ell}$ with the vector spaces $\mathbb{F}_q^n$ and $\mathbb{F}_q^\ell$, respectively, and measure their dimensions in this sense. Therefore, to clarify, we include $\mathbb{F}_q$ as a subscript in dim.

Applying the rank-nullity theorem to $\phi_{\mathrm{R}}|_{\mathcal{L}_{\mathrm{R}}(e_1, e_2)} : \mathcal{L}_{\mathrm{R}}(e_1, e_2) \to \mathcal{L}'_{\mathrm{R}}(e'_1, e'_2)$, we obtain

$$\begin{aligned}
&\dim_{\mathbb{F}_q} \ker(\phi_{\mathrm{R}}|_{\mathcal{L}_{\mathrm{R}}(e_1, e_2)}) \\
&= \dim_{\mathbb{F}_q} \mathcal{L}_{\mathrm{R}}(e_1, e_2) - \dim_{\mathbb{F}_q} \phi_{\mathrm{R}}(\mathcal{L}_{\mathrm{R}}(e_1, e_2)) \\
&\geq n - \ell.
\end{aligned} \tag{9}$$

Eq. (9) is given by Eqs. (7) and (8). Furthermore,

$$\begin{aligned}
&\ker(\phi_{\mathrm{R}}|_{\mathcal{L}_{\mathrm{R}}(e_1, e_2)}) \\
&= \{x \in \mathcal{L}_{\mathrm{R}}(e_1, e_2) \colon (\phi_{\mathrm{R}}|_{\mathcal{L}_{\mathrm{R}}(e_1, e_2)})(x) = \mathbf{0}\} \\
&= \{x \in \mathcal{L}_{\mathrm{R}}(e_1, e_2) \colon \phi_{\mathrm{R}}(x) = \mathbf{0}\} \\
&= \{x \in \mathbb{F}_{q^n}^3 \colon x \in \mathcal{L}_{\mathrm{R}}(e_1, e_2) \wedge \phi_{\mathrm{R}}(x) = \mathbf{0}\} \\
&= \mathcal{L}_{\mathrm{R}}(e_1, e_2) \cap \ker \phi_{\mathrm{R}}.
\end{aligned}$$

Hence, we obtain

$$\dim_{\mathbb{F}_q}(\mathcal{L}_{\mathrm{R}}(e_1, e_2) \cap \ker \phi_{\mathrm{R}}) \geq n - \ell,$$

which implies that

$$|\mathcal{L}_{\mathrm{R}}(e_1, e_2) \cap \ker \phi_{\mathrm{R}}| \geq q^{n-\ell}. \tag{10}$$

Now, observe that

$$\bigcup_{(e_1, e_2)} (\mathcal{L}_{\mathrm{R}}(e_1, e_2) \cap \ker \phi_{\mathrm{R}}) \subseteq \ker \phi_{\mathrm{R}}, \tag{11}$$

and for distinct $(e_1, e_2), (\widetilde{e}_1, \widetilde{e}_2)$,

$$\mathcal{L}_{\mathrm{R}}(e_1, e_2) \cap \mathcal{L}_{\mathrm{R}}(\widetilde{e}_1, \widetilde{e}_2) = \{\mathbf{0}\}. \tag{12}$$

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Indeed, if

$$(-e_1 t, -e_2 t, t) = (-\widetilde{e}_1 s, -\widetilde{e}_2 s, s)$$

holds and both sides are nonzero vectors, then the third component gives $t = s \neq 0$, and the first two components imply

$$e_1 = \widetilde{e}_1, \qquad e_2 = \widetilde{e}_2.$$

Using the distributive law in set operations, it holds that

$$\begin{aligned}
&\bigcup_{(e_1,e_2)} (\mathcal{L}_{\mathrm{R}}(e_1,e_2) \cap \ker \phi_{\mathrm{R}}) \\
&= \bigcup_{(e_1,e_2)} (\{\mathbf{0}\} \cup (\mathcal{L}_{\mathrm{R}}(e_1,e_2) \setminus \{\mathbf{0}\}) \cap \ker \phi_{\mathrm{R}}) \\
&= \bigcup_{(e_1,e_2)} ((\{\mathbf{0}\} \cap \ker \phi_{\mathrm{R}}) \cup (\mathcal{L}_{\mathrm{R}}(e_1,e_2) \setminus \{\mathbf{0}\}) \cap \ker \phi_{\mathrm{R}}) \\
&= \bigcup_{(e_1,e_2)} (\{\mathbf{0}\} \cup (\mathcal{L}_{\mathrm{R}}(e_1,e_2) \setminus \{\mathbf{0}\}) \cap \ker \phi_{\mathrm{R}}) \\
&= \{\mathbf{0}\} \cup \bigcup_{(e_1,e_2)} ((\mathcal{L}_{\mathrm{R}}(e_1,e_2) \setminus \{\mathbf{0}\}) \cap \ker \phi_{\mathrm{R}}). \qquad (13)
\end{aligned}$$

Thus, Eqs. (12) and (13) imply

$$\begin{aligned}
&\left| \bigcup_{(e_1,e_2)} (\mathcal{L}_{\mathrm{R}}(e_1,e_2) \cap \ker \phi_{\mathrm{R}}) \right| \\
&= \left| \{\mathbf{0}\} \cup \bigcup_{(e_1,e_2)} ((\mathcal{L}_{\mathrm{R}}(e_1,e_2) \cap \ker \phi_{\mathrm{R}}) \setminus \{\mathbf{0}\}) \right| \\
&= 1 + \sum_{(e_1,e_2)} (|\mathcal{L}_{\mathrm{R}}(e_1,e_2) \cap \ker \phi_{\mathrm{R}}| - 1). \qquad (14)
\end{aligned}$$

Therefore, Eqs. (10), (11) and (14) imply that

$$\begin{aligned}
|\ker \phi_{\mathrm{R}}| &\geq \left| \bigcup_{(e_1,e_2)} (\mathcal{L}_{\mathrm{R}}(e_1,e_2) \cap \ker \phi_{\mathrm{R}}) \right| \\
&= 1 + \sum_{(e_1,e_2)} (|\mathcal{L}_{\mathrm{R}}(e_1,e_2) \cap \ker \phi_{\mathrm{R}}| - 1) \\
&\geq 1 + q^{2n} \left(q^{n-\ell} - 1 \right) \\
&= q^{3n-\ell} - q^{2n} + 1.
\end{aligned}$$

Since $n > \ell$ and $q \geq 2$,

$$q^{3n-\ell} - q^{2n} + 1 > q^{3n-\ell-1}.$$

Because $\ker \phi_{\mathrm{R}}$ is an $\mathbb{F}_q$-linear subspace, its cardinality is an integral power of q, and hence

$$|\ker \phi_{\mathrm{R}}| \geq q^{3n-\ell}. \qquad (15)$$

Applying the rank-nullity theorem to $\phi_{\mathrm{R}} : \mathbb{F}_{q^n}^3 \to \mathbb{F}_{q^\ell}^3$, it is given by Eq. (15) that

$$\begin{aligned}
|\operatorname{Im} \phi_{\mathrm{R}}| &= \frac{|\mathbb{F}_{q^n}^3|}{|\ker \phi_{\mathrm{R}}|} \\
&\leq \frac{q^{3n}}{q^{3n-\ell}} = q^{\ell}. \qquad (16)
\end{aligned}$$

Here, from Eq. (5), it holds that $\operatorname{Supp}\big(K'_{\mathrm{R}} \mid S = s\big) \subseteq \operatorname{Im} \phi_{\mathrm{R}}$. (We note that we fixed a seed $S = s$.) Therefore, from Eq. (16), we obtain that

$$|\operatorname{Supp}\big(K'_{\mathrm{R}} \mid S = s\big)| \leq |\operatorname{Im} \phi_{\mathrm{R}}| \leq q^{\ell}. \qquad (17)$$

By Eq. (17), we have

$$\begin{aligned}
&\Delta\left((K'_{\mathrm{R}}, S); (\widehat{K_{\mathrm{R}}}, S)\right) \\
&= \mathbb{E}_{s \sim S}\left[\Delta\left((K'_{\mathrm{R}} \mid S = s); (\widehat{K_{\mathrm{R}}} \mid S = s)\right)\right] \\
&\geq \mathbb{E}_{s \sim S}\left[1 - \frac{|\operatorname{Supp}\big(K'_{\mathrm{R}} \mid S = s\big)|}{|\mathbb{F}_{q^\ell}^3|}\right] \\
&\geq \mathbb{E}_{s \sim S}\left[1 - \frac{q^{\ell}}{q^{3\ell}}\right] \\
&= 1 - q^{-2\ell},
\end{aligned}$$

completing the proof. Here, note that $\widehat{K_{\mathrm{R}}}$ and S are independent, and that $\widehat{K_{\mathrm{R}}}$, without conditioned on $\widehat{K_{\mathrm{S}}}$, is identically distributed on $\mathbb{F}_{q^\ell}^3$. □

5. Correcting the Construction of CSS 2025

5.1 Correction Terms

Section 4 shows that purely local linear transformations cannot simultaneously preserve the algebraic relation of the underlying A^2-code and transform the initial keys into an ideal uniformly distributed key pair. Our construction therefore does not rely on local extraction alone. Instead, it combines component-wise linear hashing with authenticated transmission of correction terms. The objective is to preserve the algebraic relation required for verification and arbitration while retaining the use of linear universal hashing for distilling weak key components.

Let $h : \mathbb{F}_q \to \mathbb{F}_r$ be a linear universal hash function. The distilled components are defined using independently sampled masks $d_1^{\mathrm{S}}, d_2^{\mathrm{S}}$ and receiver-side correction terms

$$d_i^{\mathrm{R}} \coloneqq h(e_i f_3) - h(e_i)h(f_3) + d_i^{\mathrm{S}},$$

for $i \in \{1, 2\}$. The term $h(e_i f_3) - h(e_i)h(f_3)$ exactly compensates for the multiplicative discrepancy introduced by component-wise linear hashing. Section 5.3 proves that the resulting keys satisfy the original algebraic relation.

5.2 Secure Transmission of Correction Terms

The description of h must be delivered consistently to the sender and the receiver. We therefore authenticate the hash seed using the A-code from weak sources (the definition and parameter are stated in Definition 9 and proposition 3) based on a weak shared secret. The correction terms may reveal information about key components that must remain hidden from one of the parties, and hence they are not sent in the clear. Instead, the arbiter establishes almost-uniform keys with the sender and the receiver using AKA from weak sources (the definition and parameter are stated in Definition 10 and proposition 4) and uses these keys for one-time

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information-theoretic authenticated encryption (the definition is stated in Definition 11) of the corresponding correction terms.

The arbiter first distributes the correlated weak source keys, authenticates a common hash function to both parties, and securely transmits the correction terms. The sender and the receiver then derive their distilled keys and use them in the tag-generation, receiver-verification, and arbitration procedures of the underlying algebraic A^2-code. The complete procedures are given in Algorithm 2 or Fig. A·1.

5.3 Correctness of the Algebraic Relation

The following lemma ensures that the keys produced by Algorithm 2 satisfy the relational expression of the A^2-code keys.

Lemma 2. *Suppose that both authenticated-encryption ciphertexts are accepted and decrypt to the correction terms generated by the arbiter. Then the distilled keys produced by Algorithm 2 satisfy*

$$e'_3 = f'_1 + e'_1 f'_3, \qquad e'_4 = f'_2 + e'_2 f'_3.$$

Proof. By construction,

$$\begin{aligned} e'_1 &= h(e_1), \\ e'_2 &= h(e_2), \\ e'_3 &= h(e_3) + d_1^{\mathrm{S}}, \\ e'_4 &= h(e_4) + d_2^{\mathrm{S}}, \\ f'_1 &= h(f_1) + d_1^{\mathrm{R}}, \\ f'_2 &= h(f_2) + d_2^{\mathrm{R}}, \\ f'_3 &= h(f_3), \end{aligned}$$

where

$$\begin{aligned} d_1^{\mathrm{R}} &= h(e_1 f_3) - h(e_1)h(f_3) + d_1^{\mathrm{S}}, \\ d_2^{\mathrm{R}} &= h(e_2 f_3) - h(e_2)h(f_3) + d_2^{\mathrm{S}}. \end{aligned}$$

Since h is linear and

$$e_3 = f_1 + e_1 f_3, e_4 = f_2 + e_2 f_3,$$

we have

$$\begin{aligned} & f'_1 + e'_1 f'_3 \\ &= h(f_1) + h(e_1 f_3) - h(e_1)h(f_3) + d_1^{\mathrm{S}} + h(e_1)h(f_3) \\ &= h(f_1 + e_1 f_3) + d_1^{\mathrm{S}} \\ &= h(e_3) + d_1^{\mathrm{S}} = e'_3. \end{aligned}$$

The same argument gives

$$f'_2 + e'_2 f'_3 = e'_4.$$

□

5.4 Security

Under suitable conditional min-entropy assumptions, the generalized leftover hash lemma guarantees that the hashed key components are close to uniform even given the relevant side information. Authentication of the hash seed prevents an active adversary from forcing inconsistent transformations, while authenticated key agreement and one-time authenticated encryption protect the correction terms. Thus, except for the failure events of these building blocks and the statistical errors introduced by randomness extraction, the corrected procedure behaves as the underlying algebraic A^2-code with uniformly distributed keys. See Theorem 2 in Appendix A.3 for the quantitative security bound.

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Appendix

A.1 Full Corrected Key-Distillation Procedure

The full corrected key-distillation procedure is shown in Algorithm 2 and Fig. A·1

Algorithm 2 Key distribution and distillation for the weak-source-key A^2-code

Require: A linear universal hash family $\mathcal{H}$

Require: A joint weak-source distribution P_K over $\mathcal{K}_\mathrm{S} \times \mathcal{K}_\mathrm{R}$

Key-distribution phase

1: The arbiter samples $(k_\mathrm{S}, k_\mathrm{R}) \sim P_K$.
2: The arbiter parses $k_\mathrm{S} = (e_1, e_2, e_3, e_4, u_1, u_2)$ and $k_\mathrm{R} = (f_1, f_2, f_3, v_1, v_2)$.
3: The arbiter sends k_S to the sender over a private authenticated channel.
4: The arbiter sends k_R to the receiver over a private authenticated channel.

Key-distillation phase

5: The arbiter samples $h \leftarrow\!\!\$\ \mathcal{H}$.
6: The arbiter sends h to the sender and the receiver using the Dodis–Wichs A-code instantiated with u_1 and v_1.
7: The arbiter samples $d_1^\mathrm{S}, d_2^\mathrm{S} \leftarrow\!\!\$\ \mathbb{F}_r$.
8: The arbiter computes $d_1^\mathrm{R} \leftarrow h(e_1 f_3) - h(e_1)h(f_3) + d_1^\mathrm{S}$,
$d_2^\mathrm{R} \leftarrow h(e_2 f_3) - h(e_2)h(f_3) + d_2^\mathrm{S}$.
9: The arbiter and the sender execute the Dodis–Wichs authenticated key-agreement protocol instantiated with u_2 and obtain a shared key k^S.
10: The arbiter and the receiver independently execute the Dodis–Wichs authenticated key-agreement protocol instantiated with v_2 and obtain a shared key k^R.
11: The arbiter sends $(d_1^\mathrm{S}, d_2^\mathrm{S})$ to the sender using information-theoretic one-time authenticated encryption under k^S.
12: The arbiter sends $(d_1^\mathrm{R}, d_2^\mathrm{R})$ to the receiver using information-theoretic one-time authenticated encryption under k^R.
13: The sender computes $e'_1 \leftarrow h(e_1)$, $e'_2 \leftarrow h(e_2)$,
$e'_3 \leftarrow h(e_3) + \widehat{d_1^\mathrm{S}}$, $e'_4 \leftarrow h(e_4) + \widehat{d_2^\mathrm{S}}$.
14: The receiver computes $f'_1 \leftarrow h(f_1) + \widehat{d_1^\mathrm{R}}$,
$f'_2 \leftarrow h(f_2) + \widehat{d_2^\mathrm{R}}$,
$f'_3 \leftarrow h(f_3)$.
15: The sender sets $k'_\mathrm{S} \leftarrow (e'_1, e'_2, e'_3, e'_4)$.
16: The receiver sets $k'_\mathrm{R} \leftarrow (f'_1, f'_2, f'_3)$.

A.2 Formal Definitions of Primitives

A.2.1 Authentication and Authenticated Key Agreement from Weak Sources

Message Authentication from Weak Sources. Dodis and Wichs showed that information-theoretic message authentication (i.e., A-code) remains possible in this setting. More precisely, they constructed a two-round protocol that enables two parties who share only a k-source to authenticate a message over a public, unauthenticated channel, even in the presence of an active adversary.

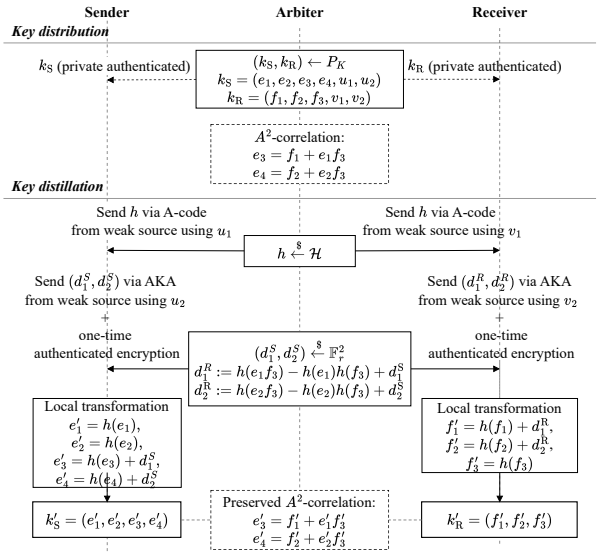


Fig. A·1 Key-distribution and key-distillation of A^2-code from weak source key

Definition 9 (A-code from weak sources)**.** *Consider a protocol between a sender and a receiver over a public unauthenticated channel in the presence of an active adversary. The sender and the receiver share a k-source over $\{0,1\}^n$. On input a source message $M_\mathrm{S} \in \{0,1\}^m$, the sender executes the protocol, and the receiver outputs*

$$M_\mathrm{R} \in \{0,1\}^m \cup \{\perp\}.$$

The protocol is called δ-secure A-code from weak sources if it satisfies the following properties for every sharing k-source over $\{0,1\}^n$:

Correctness. *If the adversary is passive, then, for every $M_\mathrm{S} \in \{0,1\}^m$,*

$$\Pr[M_\mathrm{R} = M_\mathrm{S}] = 1.$$

Security. *For every $M_\mathrm{S} \in \{0,1\}^m$ and every active adversarial strategy,*

$$\Pr[M_\mathrm{R} \notin \{M_\mathrm{S}, \perp\}] \le \delta.$$

Proposition 3 summarizes the authentication guarantee that will be used in our construction.

Proposition 3 (Security of A-code from weak sources [3])**.** *For all integers $n \ge k$ and m, and every $\delta > 0$, there exists an efficient δ-secure A-code from weak sources, provided that*

$$k > \mathcal{O}\Big(\log^2 n + \log^2(1/\delta) + \log m\Big).$$

Dodis and Wichs also showed that two parties sharing only a weak secret can agree on an almost-uniform key over a public unauthenticated channel. Unlike the weak source message authentication protocol described above, this protocol produces a shared secret key rather than authenticating a prescribed public message. Its security guarantees both privacy of the derived key and authenticity of the agreement

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in the presence of an active adversary.

Authenticated Key Agreement from weak sources. Dodis and Wichs constructed an authenticated key-agreement (AKA) protocol over a public unauthenticated channel from a weak source shared by two parties, using message authentication from weak sources as a building block. In this paper, we call this AKA from weak sources. Although the parties in communication protocols are generally referred to as the sender and the receiver throughout this paper, we use Alice and Bob in this definition because the two parties play symmetric roles. AKA from weak sources is defined as follows.

Definition 10 (AKA from weak sources)**.** *Consider a protocol between Alice and Bob over a public unauthenticated channel in the presence of an active adversary. Alice and Bob share a k-source W over $\{0,1\}^n$ and output $R_A, R_B \in \{0,1\}^\ell \cup \{\perp\}$, respectively. Let T denote the transcript observed by the adversary. The protocol is said to achieve authenticated key agreement from weak sources if it satisfies the following three properties for every k-source W over $\{0,1\}^n$ and every active adversarial strategy:*

Correctness. *If the adversary is passive, then*

$$\Pr[R_A = R_B \neq \perp] = 1.$$

Key privacy. *Conditioned on $R_A \neq \perp$, Alice's key is close to uniform and independent of the adversary's view:*

$$\Delta\left((T, R_A) \mid R_A \neq \perp; (T, U_\ell) \mid R_A \neq \perp\right) \leq \epsilon,$$

where U_ℓ is uniform over $\{0,1\}^\ell$ and independent of T.

Key authenticity. *The probability that Bob outputs a key different from Alice's key is bounded by*

$$\Pr[R_B \neq \perp \wedge R_A \neq R_B] \leq \delta.$$

ϵ is called the key privacy error *and δ is called the* key authenticity-error.

Proposition 4 summarizes the guarantee used in this work.

Proposition 4 (Security of AKA from weak sources [3])**.** *For every constant $\alpha > 0$, all integers $n \geq k$, and all $\epsilon, \delta > 0$, there exists an efficient two-round AKA from weak sources. The protocol has key privacy error at most epsilon, key authenticity error at most δ, and output length*

$$\ell = (1-\alpha)k - \mathcal{O}\Big(\log^2(1/\delta) + \log^2(1/\epsilon)\Big).$$

Its communication complexity is

$$\mathcal{O}\Big(\log^2 n + \log^2(1/\delta) + \log^2(1/\epsilon)\Big).$$

A.2.2 Information-Theoretic One-Time Authenticated Encryption

Following the information-theoretic formulation of authenticated encryption by Shikata et al. [9], we use the following one-time specialization of perfect secrecy and plaintext integrity.

Definition 11 (One-time unconditionally secure authenticated encryption)**.** *A one-time authenticated encryption scheme consists of algorithms*

$$\mathsf{Enc} : \mathcal{K} \times \mathcal{M} \to \mathcal{C}, \mathsf{Dec} : \mathcal{K} \times \mathcal{C} \to \mathcal{M} \cup \{\perp\}.$$

Let K be uniformly distributed over $\mathcal{K}$. The scheme is δ-secure if it satisfies the following properties.

Correctness *For every $m \in \mathcal{M}$,*

$$\Pr[\mathsf{Dec}(K, \mathsf{Enc}(K, m)) = m] = 1.$$

Perfect secrecy *For every distinct $m_0, m_1 \in \mathcal{M}$,*

$$\mathsf{Enc}(K, m_0) \equiv \mathsf{Enc}(K, m_1).$$

One-time authenticity *For every $m \in \mathcal{M}$ and every $c, c' \in \mathcal{C}$ with $c' \neq c$ and $c \in \mathrm{Supp}(\mathsf{Enc}(K, m))$,*

$$\Pr[\mathsf{Dec}(K, c') \notin \{\perp, m\} \mid \mathsf{Enc}(K, m) = c] \leq \delta.$$

A δ-secure information-theoretic one-time authenticated encryption can be constructed information-theoretically by combining a one-time pad with a strongly universal hash-based δ-secure A-code using independent key components. When the key space is $\mathcal{K}$ and the message space is $\mathcal{M}$, The ciphertext length is $|\mathcal{C}|$ is $\log|\mathcal{C}| = 2\log|\mathcal{M}| + \log|\mathcal{K}|$.

A.3 The Security of Corrected Protocol

Let $q = 2^\nu$ and $r = 2^\rho$ for positive integers ν and ρ with $\rho < \nu$. Let $\mathcal{H}$ be an $\mathbb{F}_2$-linear universal hash family from $\mathbb{F}_q$ to $\mathbb{F}_r$.

We assume that E_1, E_2, F_1, F_2, F_3 are mutually independent k-sources over $\mathbb{F}_q$, and define

$$E_3 = F_1 + E_1 F_3, \qquad E_4 = F_2 + E_2 F_3.$$

The auxiliary weak sources used for message authentication and authenticated key agreement (u_1, u_2, v_1, v_2) are assumed to satisfy the requirements of Propositions 3 and 4, respectively.

Theorem 2 (Security of the Corrected Protocol)**.** *Let $q = 2^\nu, r = 2^\rho$ for some positive integers ν, ρ satisfying $\rho < \nu$. Suppose that*

- *The authentication error of an A-code from a weak source is δ_{AC},*
- *The authenticity error of an AKA from a weak source is δ_{KA},*
- *The key privacy error of an AKA from a weak source is ϵ_{KA},*
- *The forgery probability of the information-theoretic one-time authenticated encryption scheme under a uniform key is δ_{AE}.*

If $3k > 2\log q$, then Algorithm 2 defines an A^2-code from weak source keys whose deception probability satisfies

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$$P_{\mathrm{D}} \leq \frac{1}{r} + 2\,\delta_{\mathrm{AC}} + 2\,\delta_{\mathrm{KA}} + 2\,\delta_{\mathrm{AE}} + 2\epsilon_{\mathrm{KA}} + \max\left\{2\epsilon_{\mathrm{KA}}, 2\epsilon_1 + \epsilon_{\mathrm{KA}}, \epsilon_2 + \epsilon_{\mathrm{KA}}\right\},$$

where

$$\epsilon_1 = \frac{1}{2}\sqrt{r\,2^{-k}},$$
$$\epsilon_2 = \frac{1}{2}\sqrt{r\,2^{-(3k-2\log q)}}.$$

Proof sketch. We give the proof sketch of Theorem 2.

Let Bad be the event that authentication of the hash function fails, authenticated key agreement fails, or one of the authenticated encryption ciphertexts is forged. By the security of the underlying primitives and the union bound,

$$\Pr[\mathsf{Bad}] \leq 2\delta_{\mathrm{AC}} + 2\delta_{\mathrm{KA}} + 2\delta_{\mathrm{AE}} + 2\epsilon_{\mathrm{KA}},$$

Conditioned on $\neg\mathsf{Bad}$, both the sender and the receiver use the same hash function and receive the correction terms computed by the arbiter. Hence, by Lemma 2, the distilled keys satisfy the algebraic relations required by the underlying A^2-code.

By Proposition 2, the hashed key components are close to uniform from the attacker's view. The statistical distance between a directly hashed k-source and a uniformly distributed component is bounded by

$$\epsilon_1 = \frac{1}{2}\sqrt{r\,2^{-k}}.$$

And the statistical distance between the component F_3' and a uniformly distributed component is bounded by

$$\epsilon_2 = \frac{1}{2}\sqrt{r\,2^{-(3k-2\log q)}}.$$

This bound follows from Proposition 2 and the conditional min-entropy bound

$$\widetilde{\mathrm{H}}_\infty(F_3 \mid E_1, E_2, E_3, E_4) \geq 3k - 2\log q.$$

The security argument then reduces each successful deception attack to predicting a key component hidden from the dishonest party. After replacing the correction-term encryption keys with uniform keys, these components are almost uniform from the relevant party's view.

For each of the five deception attacks, conditioned on the attacker's view, successful deception requires predicting a value of another party's key component. For the ideal key, the success probability of any attack is at most $1/r$. Combining these bounds with $\Pr[\mathsf{Bad}]$ and the statistical distance from the uniform distribution proves the claimed bound. □

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